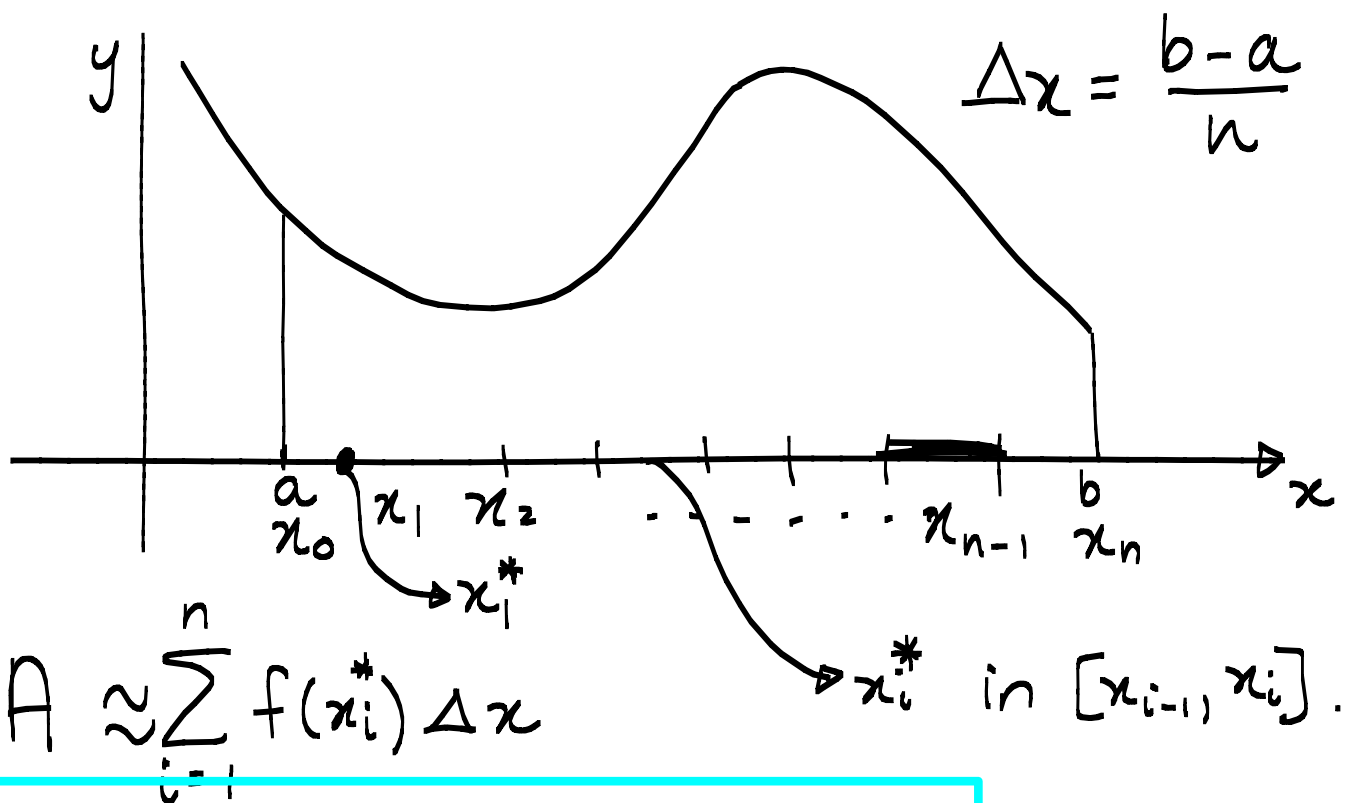


Unit 5.3 The definite integral p 371

Note Title

2014/04/20



$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

*

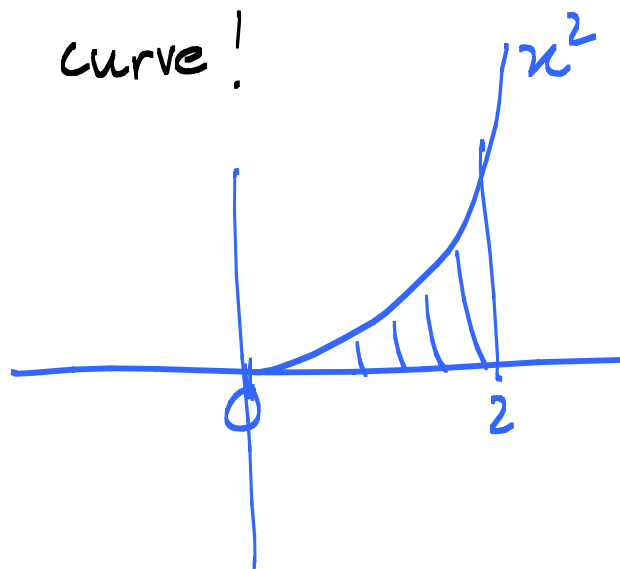
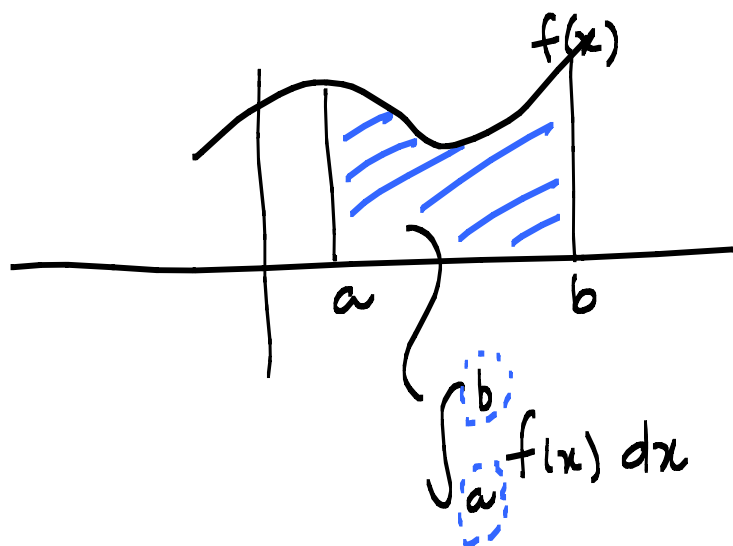
limits of integration

Integrand.

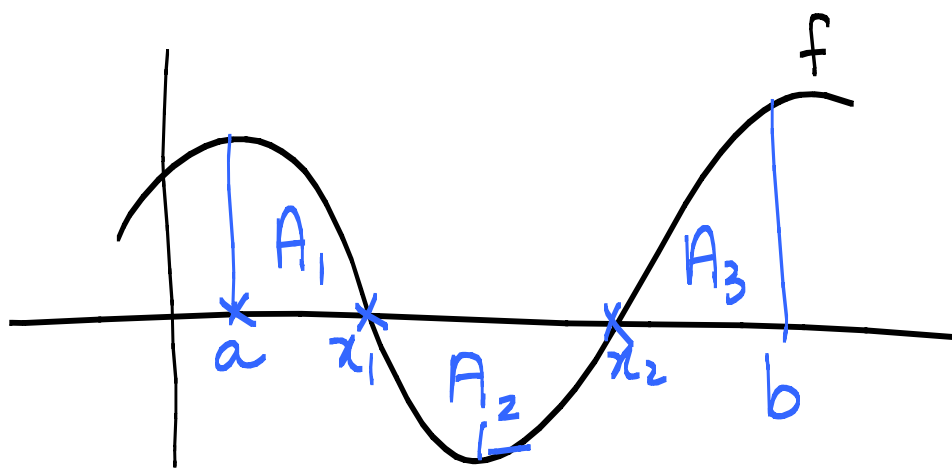
Riemann Sum.

Integral.

So: If $f(x) \geq 0$ then we have the area under the curve!



What happens when f negative?



$$\int_a^b f dx = \text{Area?}$$

$$\int_a^{x_1} f dx = A_1$$

$$\int_{x_2}^b f dx = A_3$$

$$\int_{x_1}^{x_2} f dx = -A_2$$

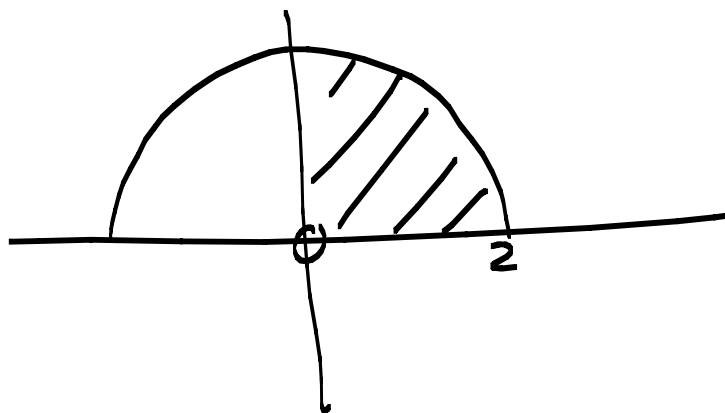
$$\text{Area} = A_1 - (-A_2) + A_3$$

Use the area to calculate

[WTW164]

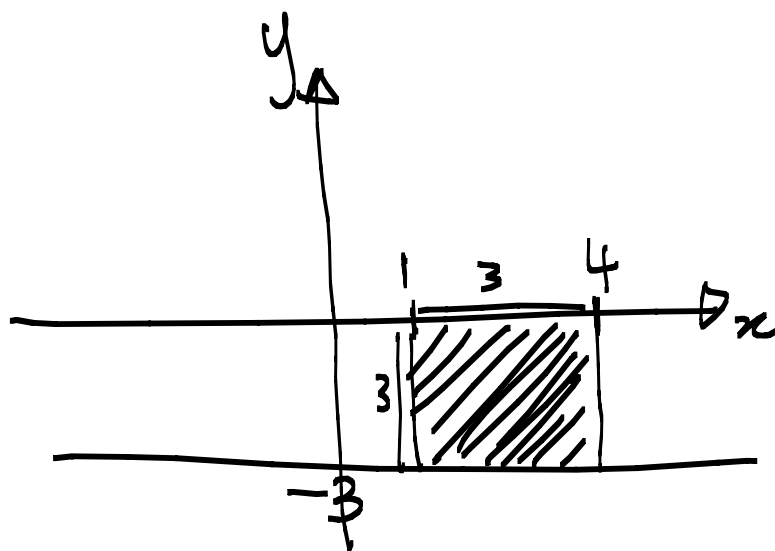
$$1. \int_0^2 \sqrt{4-x^2} dx$$

$$= \frac{1}{4} (\pi (2)^2) = \pi$$



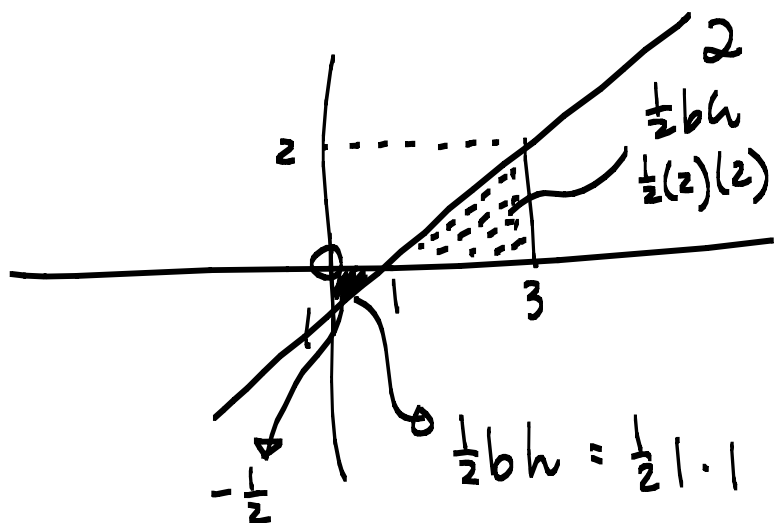
$$2. \int_1^4 -3 dx$$

$$= -9$$



$$3. \int_0^3 (x-1) dx$$

$$= 2 - \frac{1}{2} = 1 \frac{1}{2}$$

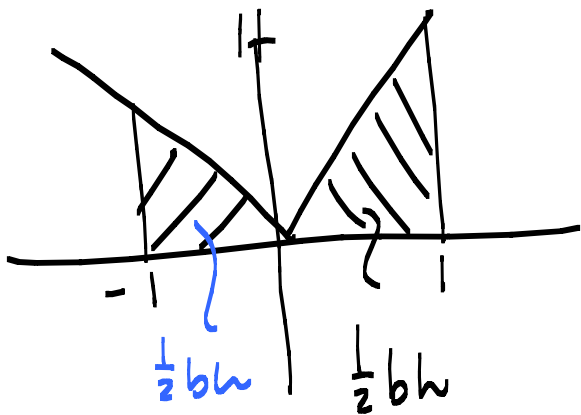


What is the area between the x-axis and the curve of $y = x - 1$ between

$$x = 0 \text{ and } x = 3. \quad \left(2 + \frac{1}{2} = 2\frac{1}{2} \right)$$

Area cannot be negative, but
Integral can be negative.

4. Find the area under the curve
of $|x|$ where x goes from -1 to 1 .



$$\int_{-1}^1 |x| dx$$

$$= \frac{1}{2} + \frac{1}{2} = 1$$

Properties of the integral p 385

1. $\int_a^b c dx = c(b-a)$
2. $\int_a^b (f(x) \pm g(x)) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$
3. $\int_a^b c f(x) dx = c \int_a^b f(x) dx$
4. $\int_a^b f(x) dx = - \int_b^a f(x) dx$
5. $\int_a^a f(x) dx = 0$